

Trash It, Punch It, Burn It – Using Virtual Reality to Support Coping with Negative Thoughts

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ABSTRACT

Negative thoughts are a widespread everyday experience. Failures to appropriately cope with negative thoughts are related to serious mental health issues, such as depression. Consequently, preventive technology-mediated everyday interventions to support coping with negative thoughts are of interest. One promising platform for mental health applications is personalized virtual reality (VR). We developed an explorative VR prototype based on personally relevant textual messages from email, messengers and alike, which trigger negative thoughts. The prototype presented these messages in VR and allowed to physically manipulate them, for example, by physically punching and trashing them. A qualitative empirical exploration (N=10) revealed a general positive shift in thoughts and emotions after using the prototype, mainly in form of increased relaxation and self-reflection. Based on this and further insights, we derive four themes for VR in mental health, touching upon the importance of personalization, immersion and focus, interaction design and embodiment, as well as, integration into everyday life.

CCS CONCEPTS

• **Human-centered computing** → Interaction design; Interaction design process and methods; Scenario-based design.

KEYWORDS

virtual reality, mental health, negative thoughts, reflection, coping, interaction design, design for wellbeing

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1 INTRODUCTION

A negative thought can be defined as “any distinct, identifiable cognitive event that is unwanted, unintended, and recurrent” ([6], p. 4). Negative thoughts affect task performance, interrupt the flow of thought and are often key to negative emotional states [6]. People

frequently experience negative thoughts [2] and since they are linked to serious mental health problems, such as depression, brief psychological interventions to support coping with them are of interest [14].

So far, a number of therapeutic interventions to better manage negative thoughts and resulting emotions exist, such as cognitive restructuring, cognitive defusion [14] or emotion regulation [12]. However, these treatments are clinical applications and require substantial training and psychotherapeutic expertise. This is a notable barrier to their preventive everyday use for better coping with mild negative thoughts. Given this, technology could offer new possibilities to support the training of coping strategies. However, while several technological applications for clinical contexts exist, this is not the case for everyday untrained use [22].

In the present paper, we focus on a technologically-mediated support for coping with negative thoughts triggered by textual messages. Textual messages, such as emails, SMS, WhatsApp-Messages, or social media comments, are omnipresent in everyday life [25]. Phenomena, such as bullying via electronic communication [24], give reason to assume that textual messages are a potent source of negative thoughts and resulting emotions.

From a technical perspective, textual messages are highly accessible, easy to store and to analyze. In the present case, we combine textual messages with Virtual Reality (VR). In general, VR seems to be a promising technological platform for clinical mental health applications, due to its immersive character, its potential for individualization and the possibility to explore different coping-strategies [1]. More specifically, we designed a prototype which allowed to physically manipulate negative textual messages in VR to better cope with them.

We start with an overview of approaches using technology to support the regulation of negative thoughts. We then describe the initial ideation, the prototype and a first qualitative exploration of the prototype’s effects. We close with some preliminary general themes derived from the exploration.

2 VIRTUAL REALITY TO REGULATE NEGATIVE THOUGHTS AND EMOTIONS

A common therapeutic approach to the treatment of negative thoughts is cognitive restructuring [14]. In cognitive restructuring, therapists and patients aim to identify negative thoughts and challenge their veracity collaboratively [13] to control these thoughts [9]. This can also be applied to negative emotions [11]. There are several technology-mediated approaches to cognitive restructuring. For example, Schmidt et al. [20] developed a game-based prevention against performance anxiety in primary school using cognitive restructuring. Conesa et al. [7] designed an app using cognitive restructuring for the treatment of chronic pain. Stamou et al. [23]

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combined VR with cognitive restructuring as part of a broader cognitive behavior therapy approach for the treatment of post-natal depression.

Because of its immersive character, VR has been successful in the treatment of clinical mental disorders, such as post-traumatic stress, phobia and pain [1]. Baghaei et al. [1] advocate individualized VR for mental health applications, because it provides possibilities for addressing various individual coping strategies. For example, explaining a problem to a virtual doppelganger in VR and then alternating to the other body to see and hear the own conversation again helped people to change perspective and showed positive psychological outcomes [16, 21]. In VR, to design for embodied cognition, i.e., the notion that cognition, body and movement are deeply intertwined, can lead to personal efficacy and self-reflectiveness due to an increased sense of presence and emotional engagement [18]. All in all, however, there is little research in HCI, which specifically addresses coping with everyday negative thoughts, even less, on approaches which offer possibilities to include personalized content [1].

To remedy this, we looked for simple ways to include personal content in VR environments. One possible solution are text messages, such as emails and WhatsApp messages. These messages are ubiquitous, easy to gather from personal communication, and – most importantly – they contain personally relevant triggers of negative thoughts. In this sense, textual messages could be the target of technology-mediated cognitive restructuring. However, Briñol et al. [4] caution that mental actions to cope with negative thoughts can be hard to implement, since they often do not work as intended [26] and sometimes even backfire, thereby amplifying the unwanted thoughts rather than to attenuate them. Instead of mental actions, they propose treating negative thoughts as physical objects, i.e., to materialize them. In two experiments, they showed that participants, who wrote down their thoughts on paper and literally threw them away, relied less on these thoughts in subsequent tasks (e.g., judgements) compared to participants who kept them. They found their results to be in line with the concept of embodied cognition, because bodily action influences people’s thoughts and feelings [5]. For example, a confident posture can magnify people’s thoughts, cognitions and emotions, while motor expressions such as gestures or postures can support appraisal processes [17]. A major advantage of treating thoughts as physical objects is that it opens up possibilities for people to do things with their thoughts and associate a meaning with these actions (e.g. throwing something away), that would not be easy to do mentally [5]. So far, only a few technologically-mediated approaches took this approach. An example is a VR-based system called “Thought Disposal” [3] for teens to reduce stress. “Thought Disposal” offers a VR environment with pre-written negative thoughts in the form of leaves on a tree. The teens can pick the negative thoughts from the tree and burn them in a fire “to get rid of them”.

In sum, due to its immersive character, VR shows a potential to support the reflection of negative thoughts by using personally-relevant content in form of textual messages (i.e., triggers), make this content “graspable” and allow to physically manipulate them in a meaningful, attenuating way (e.g., by pushing them away).

3 CONCEPT AND PROTOTYPE

3.1 Ideation

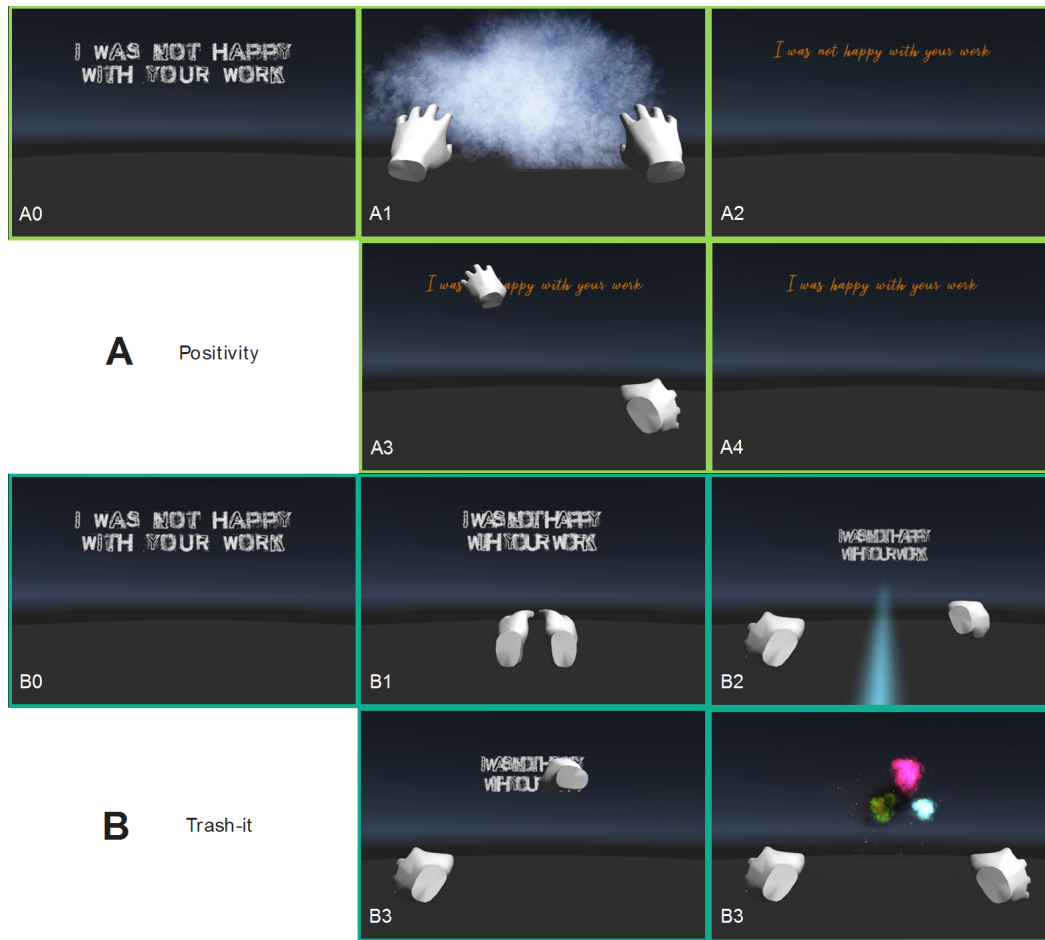
We took a Research through Design (RtD) approach (e.g., [10, 27, 28]) to explore VR-based ways to cope with negative thoughts triggered by text messages. At the heart of RtD are artefacts (e.g. prototypes or products) grounded in models and theories from behavioral science (“true knowledge”), technical possibilities (“how knowledge”) and qualitative research (“real knowledge”), such as own data or published ethnographies. Making these artefacts, clarifies possibilities and contextualizes and materializes the knowledge.

For ideation, we began with emails volunteered by the second author. Each email had provoked negative thoughts. For the time being, we excluded the rather technical question of how negative messages could be extracted from a corpus of daily message received through email or other communication services. One possibility is forms of sentiment analysis [15]. Another possibility might be a tagging system, for example, implemented as an add-on of a mail program. The user could simply tag emails in categories of negativity and the respective intensity for further use by the system.

We discussed initial ideas and found two concepts to be interesting, called *Trash-it / Punch-it / Burn-it* (short *Trash-it*) and *Positivity*. Both concepts allow for the physical manipulation of messages in VR. *Trash-it* offers for different active interactions to get rid of the message – physically and, hence, mentally. *Positivity* is based on magically transforming the messages content and appearance, by changing its color and font, and by providing the users with the means to strike out negative words. Both concepts need physical (hand) gestures to interact with the messages in VR. This corresponds to the approach of Briñol et al. [4], who suggested to treat thoughts as objects that can be physically altered and manipulated. Yet, both concepts exhibit crucial differences, which we found promising to further explore in the development and evaluation of the prototype: while *Trash-it* is more active and bodily, *Positivity* impacts the semantic level of the message, but through bodily gestures.

3.2 Prototype

We implemented the two concepts in a VR prototype as a smartphone application in combination with a cardboard VR headset. The virtual environment is kept minimalistic to make the text stand out and to support focusing on the message and not on environmental elements. Both concepts were implemented as short (5 minutes) interactive sessions, which followed a predefined script. At the beginning of each session, the personally relevant negative message slowly approaches until it reaches the user and fills her or his field of view. The rather low-tech setup of this prototype as a smartphone app in combination with a cardboard VR headset excluded tracking hands. Note, that we deliberately selected this setup since it fits the idea of a brief everyday intervention, which can be easily carried out without technical barriers. Since the hands of users cannot be tracked in such a simple technological setup, we presented the implied physical manipulations through virtual hands with a lowered opacity for three times. Then the hands turned to full opacity to signal the user to perform this gesture themselves in reality. This created an experience as if the hands of the users were tracked, and that the gesture performed led to the particular consequences, such

Figure 1: Prototype of *Positivity* and *Trash-It*

as change of colors and font. The implementation of both concepts in the prototype is summarized in Figure 1 below.

In *Positivity*, the user first transforms (A1) the font of the message and its color from white into orange (A2) by circling her or his hands like a magician. Afterwards, the users can cross out negative words (A3) to turn the meaning of the message from a negative into a positive one (A4).

In *Trash-it*, users can first crumple the message similarly to a real piece of paper (B1). Subsequently, the users are asked to punch the message (B2) and as if hit, it moves away. The final interaction is to stab the message with a lance of light (B3). The message explodes and disappears (B4).

4 EMPIRICAL EXPLORATION

4.1 Participants and procedure

Ten individuals participated in the empirical exploration (8 males, median age = 28, min = 21, max = 64). They had diverse backgrounds, such as engineer, child care worker or student. They received no compensation for participation.

Prior to the study, each participant was asked to forward two e-mails which triggered negative thoughts. We arbitrarily assigned one email either to *Trash-It* and *Positivity*. For *Positivity*, we also chose which words would be crossed out in step A3 to change the meaning of the message to the more positive whenever possible. We conducted the exploration at the home of each participant. Half of the participants went through *Positivity* first followed by *Trash-It*, the other half did this vice versa.

After using the prototype, we conducted a semi-structured interview. We first asked about the general impression and shifts in thoughts and emotions regarding the message and its sender. Additionally, we explored differences between both scenarios and how the gestures and their physical nature as well as their order had been experienced. Finally, we asked about the use of VR and if the participants could imagine using such a system in everyday routines and why (not).

All interviews were recorded, transcribed, and coded. In a first step, we used Initial Coding as a method to examine and compare the qualitative data, employing In Vivo Coding [19]. In a second step, we synthesized the emerging codes into different categories, each with their own number of sub-codes, if necessary.

4.2 Results

Several themes emerged out of the qualitative data collected in the interviews: shifts in thoughts and emotions regarding the messages; feedback about the particular prototypes; gestures and embodied cognition; VR; and integration into everyday life. In the following, we report findings for each category and their relationship to each other. All provided quotes were translated from German into English.

General Shifts in Thoughts and Emotion: All participants believed that the VR intervention had at least a small positive effect on the perception of the negative message. All participants felt more relaxed afterwards: (P05) “Regarding the second [message, I’m more relaxed] because now it has become funny, because [of *Positivity*]. And regarding the first [message, I’m more relaxed] because I reflected again [...] and then accepted it a little bit more. This way, I’ve got rid of my initial anger.” The self-reflective effect of the intervention was further mentioned by eight participants: (P09) “When you suddenly get a new perspective on something that you’re angry about because it seems not so bad anymore, then you inevitably ask yourself: [...] ‘Why did it upset me so much?’ So, I would agree that this makes me calmer.” In sum, all participants believed that at least one out of the two prototypes had a soothing, relaxing, self-reflective effect.

Prototypes: Four out of ten participants found *Trash-It* helpful to down-regulate negative emotions. However, for one participant *Trash-It* even had the potential to further exacerbate aggression towards the sender: (P09) “Through this aggressive crumpling or stabbing, I feel more potential for aggression towards the sender.” Thus, *Trash-It* was more polarizing. Especially the punching interaction was perceived negatively by three participants because of its fierceness. Two participants additionally found *Trash-It* not to have an effect because it does not fit their personality. P06 said: “[It did not have a positive effect because] I am not the kind of person that escalates like that. [...] Crumpling, boxing and stabbing is something I find rather disconcerting.” On the other hand, three participants perceived crumpling their message to be a positive metaphor and destroying the message by stabbing it had a resolving effect for two participants in the sense of “getting rid of it”.

Regarding *Positivity*, five participants reported positive cognitive effects, such as providing an opportunity for re-evaluating; making negative thoughts feel less personal; changing thoughts to the positive and being more objective about the problem at hand. Here, especially changing the meaning of a message by crossing out words was perceived as positive by six participants: (P06) “The effect [of *Positivity* compared to *Trash-It*] is bigger for me because it changed this message for the better.” In sum, while not everybody found each intervention personally fitting, the offered prototypes were clearly expressive and meaningful for coping with negative thoughts.

Gestures and Embodied Cognition: All participants perceived using gestures to “physically” interact with messages and the possible effect on cognition as positive. Four participants felt more self-determined by it. Three participants found it positive to be active: (P06) “I think that’s good because you are slightly more active in this process and therefore feel as if you are actively doing something by

yourself and it’s [...] better than just thinking about it theoretically.” P03 further mentioned the effect the body has on the mind: “In case of anger release, physical things are often involved. And I believe it is important, if you’re intensively angry. [...] Because the body can process it then in the right way.” These results are in line with the notion of Briñol et al. [4] to treat thoughts as physical objects.

Virtual Reality: The overall response regarding the use of VR was positive. Three participants found the immersive character of VR positive. Four participants found that VR provided a distraction-free setting: (P06) “I think [the use of VR] is good, because you can concentrate completely on it and only think about [its content]. And don’t associate external influences with it, so I think it’s very helpful.” What P06 further liked about VR was that it provides a different setting: “I think [the use of VR] was good, because you put yourself [...] mentally into another setting, so that you might get out of an existing thought pattern for a while. Because you [...] are then perhaps more open to the fact that something different is happening.” Other positive aspects of VR that were mentioned by one participant each were that the interaction is more graspable in VR; the interaction feels more direct; and VR separates the users from their usual environment in which they usually read and answer messages. In sum, VR seems an appropriate platform for self-regulatory interventions and reflection – even in form of technically simple implementations. This is mainly due to the immersion it provides.

Integration into Everyday Life: Six out of ten participants could imagine to use such a VR system in their everyday life, five of them immediately after receiving the message as described by P09: “I could imagine that I put on [the VR headset] and see if it changes my mood for the better. Maybe I simply read the content differently and then I can immediately answer it how I want [without being affected by negative emotions].” Thus, the VR system would be used to avoid lingering negative thoughts triggered by such messages and to become able to answer factually in an appropriate way. Furthermore, P10 imagined to use this kind of system to write down problems and virtually interact with them similarly to textual messages. Eventually, P01 imagined to cope with written down tasks at work with such a VR system: “Maybe you could use [VR] depending on how you feel about [the task]. So, either it was something really annoying and I want to burn it or box it. Or it was something cool that I want to remember for a longer time, then I pin it somewhere that I can see it for a longer time. So, I am proud of the fact that I somehow managed to complete it.”

4.3 Summary

In sum, the exploration of the concepts provided promising results towards a positive shift in negative thoughts. All ten participants felt more relaxed afterwards and eight reported a self-reflective effect. Interacting physically with textual messages via gestures was perceived positively as well as VR and its immersive and focusing character. Most of the participants could imagine to use such a system in their everyday life as an immediate intervention after receiving a negative message. Not surprisingly, the two different concepts had different effects: *Positivity* provided a new, fresh perspective and *Trash-It* instead tempered irritation. The latter polarized more, as some participants found it too forceful.

As a matter of fact, the results highlight the potential positive effects of VR interventions for better self-regulation and reflection, which make use of the immersive character of VR in combination with the notion of embodied cognition [18] and personalized content [1]. They are promising regarding the integration of such interventions into everyday life of people outside the clinical context. Furthermore, the results are in line with the notion of Briñol et al. [4] to treat thoughts as physical objects to decrease their impact.

5 DISCUSSION

We explored if and how interventions in VR can support coping with negative thoughts and emotions triggered by everyday textual messages. In general, we found that such VR interventions can lead to a positive shift in negative thoughts triggered by textual messages. Furthermore, aspects of embodied cognition and VR were seen mainly as positive. The majority of participants could imagine to use such a system in their everyday life.

Overall, we found that a simple VR-based intervention which offers to physically manipulate negative textual messages can support coping with negative thoughts and emotions. While just a beginning, it seems promising to further engage in designing accessible, technology-mediated interventions to cope with negative thoughts based on VR.

In the following, we summarize our experiences and insight with designing the prototype and its empirical exploration into four additional themes:

1. **Personalization:** Personalized content makes reflection on negative emotions more accessible and self-relevant. The use of textual data for that matter offers many potentials, because from a technical perspective it can be processed easily and fast. From a design perspective, text allows different typographical and therefore various possible forms of manipulations.
2. **Immersion and Focus:** Using an immersive medium, such as VR, focuses users on the messages and their manipulation without distraction. This way, users can fully engage with the intervention, which seems key to its effects. It further provides a different, highly artificial setting, which might be helpful to escape previous thought patterns and to rethink individual responses to negative messages and thoughts in general.
3. **Interaction Design and Embodied Cognition:** Since bodily action can influence people's thoughts and emotions [8], physical interactions such as gestures can have effects on thoughts and emotions. In our exploration, physically acting out was concordantly perceived as positive, and for some explicitly related to feeling more self-determined and actively coping.
4. **Integration into Everyday Life:** Generally, there is a demand to integrate such a system into daily routines, for everyday life problems but also in working contexts. An aspect not yet addressed is the time-point for an intervention. For example, should the system be used immediately after receiving the message which triggers negative thoughts and emotions or with distance to it?

Of course, these are only preliminary implication. They need to be expanded and further explored in the future to determine how to design technology-mediated everyday interventions to cope with negative thoughts and under which circumstances these interventions can positively impact the wellbeing of their users.

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